Post-training course: Q&A 1

RLHFBOOK.COM/COURSE

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Question-first templates from the issue tracker, Discord, and YouTube comments.

Course orientation

What prerequisites should a learner cover before this course?

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This course is approximately designed for early AI PhD/MS students. Traditional coursework path to this course:

- Basics of language modeling / NLP
- Basic ML (e.g. intro to AI course and intro to ML course)

With AI agents, all the material is very accessible. I encourage people to track down rabbit holes, skip videos, go out of order, and chase what excites them.

Related courses: David Silver's RL Course (2015), UC Berkeley Deep RL (2023), Sutton & Barto RL Book

Source: @subhu2005 on YouTube (Lecture 1)

Data & instruction tuning

For synthetic SFT data, is one teacher model better than many?

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- OpenThoughts3: QwQ-32B teacher, 1.2M examples (Guha et al., 2025); release notes.
- Similar empirical pattern in Olmo 3 reasoning SFT (Olmo et al., 2025).
- Working note: Qwen3-32B / QwQ-32B-era teachers still look unusually strong.
- Open question: why don't bigger, newer, smarter models transfer better?

Source: @mufeezamjad790 on YouTube (Lecture 2)

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How important is balancing domains in an SFT mix?

To making a general model, very, but the solutions aren't too surprising. Harder tasks get more tokens.

Source: @jeromeeusebius on YouTube (Lecture 2)

Reward models & reward design

Is there a unified benchmark across pairwise RMs, ORMs, PRMs, and generative RMs?

Is there a unified benchmark across pairwise RMs, ORMs, PRMs, and generative RMs?

Context: the question asks whether one labeled dataset could support comparison data, per-token labels, and outcome labels for apples-to-apples RM evaluation.

Answer: No. Fragmentation of the community makes progress more diffuse. This is okay, and a normal evolution.

Source: @jeromeeusebius on YouTube (Lecture 2)

Why choose token-level rewards over sequence-level rewards?

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Careful: two different meanings.

- **Reward granularity:** final-answer reward vs step/token rewards (PRM, KL, format rewards) (Lightman et al., 2024).
- **Loss granularity:** average by sequence vs average by active tokens; DAPO uses token-level policy-gradient loss (Yu & others, 2025).

```
# Sequence-normalized: each completion has
equal weight
loss = ((pg_loss * mask).sum(dim=1) /
mask.sum(dim=1)).mean()

# Token-normalized: each generated token has
```

Why token-level can help:

- local signals: KL, format rewards, PRMs, partial credit
- long-CoT RL: more gradient on the tokens actually sampled
- batching: normalize by active tokens, not examples
- DAPO: token-level loss + Clip-Higher + dynamic sampling for reasoning RL (Yu & others, 2025)

Counterpoint: if the reward is only final-answer correctness, sequence-level

Source: @dejavukun on YouTube (Lecture 3)

RL math & notation

Is GAE the only way to compute the advantage in PPO?

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The PPO clipped objective on the cheatsheet:

$$J(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\hat{A}_t,\ \text{clip}(\rho_t(\theta),\ 1-\varepsilon,\ 1+\varepsilon)\hat{A}_t\right)\right]$$

No! But I don't know of anyone using something else, or a raw value model. GAE works.

Source: issue #382 on GitHub

What is the shape of ρ?

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

Source: @JGLambourne on YouTube (Lecture 3)

What is the shape of ρ?

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

- $\pi(\cdot \mid s_t)$ is vocab-shaped: `[B, T, V]`
- $\pi(a_t \mid s_t)$ is gathered at sampled tokens: `[B, T]`
- ρ_t is scalar per token; rollout tensor is `[B, T]`

```
# logits/logprobs over vocab: [B, T, V]
old_logps_all =
old_policy.log_softmax(logits_old, dim=-1)
new_logps_all = policy.log_softmax(logits_new,
dim=-1)

# sampled token ids: [B, T]
actions = input_ids[:, 1:]

# gather only the probabilities of the sampled
tokens: [B, T]
old_logps = old_logps_all.gather(-1,
actions.unsqueeze(-1)).squeeze(-1)
new_logps = new_logps_all.gather(-1,
actions.unsqueeze(-1)).squeeze(-1)

# importance ratio per sampled token: [B, T]
rho = torch.exp(new_logps - old_logps)
```

Source: @JGLambourne on YouTube (Lecture 3)

PPO implementation & debugging

Why does PPO use `torch.max` for both policy and value losses?

Why does PPO use `torch.max` for both policy and value losses?

```
# PPO minimizes a negative policy loss
pg_loss_unclipped = -advantages * ratio
pg_loss_clipped = -advantages * torch.clamp(
    ratio, 1.0 - eps, 1.0 + eps
)

# choose the more conservative / worse loss
pg_loss = torch.max(pg_loss_unclipped,
pg_loss_clipped)
```

`torch.max` matters because PPO is written as a **loss to minimize**.

It selects the update that gives less improvement after clipping.

```
# value clipping: do not let critic jump too
far
v_clip = old_values + torch.clamp(
    v_pred - old_values, -vf_eps, vf_eps
)

vf_unclipped = 0.5 * (v_pred - targets) ** 2
vf_clipped = 0.5 * (v_clip - targets) ** 2

# optimize the worse value error
vf_loss = torch.max(vf_unclipped, vf_clipped)
```

Policy clipping limits policy drift.

Value clipping limits critic drift.

Same pattern: compare unclipped vs clipped, train on the worse one.

Source: @jacoboromerodiaz6967 on YouTube (Lecture 4)

Human feedback operations

How should data-annotation companies stay valuable as synthetic data gets simpler?

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A big shift underway from “grunt” low-skill work to expert domains like law, finance, healthcare, consulting, which solves a lot of the data company problems.

Context from the reader: this asks about the aside in Chapter 11 on companies that vet data annotators.

Source: @lorenzogaletto4442 on YouTube (Lecture 4)

What comes next?

Catch up

- Course page
- Welcome video
- Lecture 1: Overview
- Lecture 2: IFT, RMs, rejection sampling
- Lecture 3: RL motivation & math
- Lecture 4: RL implementation & practice

Upcoming lectures

- **Lecture 5:** Reasoning training & inference-time scaling
- **Lecture 6:** Direct alignment algorithms
- **Later:** Preferences, preference data, synthetic data, tools, evaluation, product, and style

Thanks for watching

Questions, comments, and future Q&A prompts:

- Course: rlhfbook.com/course
- Discord: discord.gg/yz5AwK4gBR
- GitHub: github.com/natolambert/rlhf-book
- Newsletter: interconnects.ai
- Contact: nathan@natolambert.com



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